

Q1 : Is there any situation that describes an unconventional route taken in your experience?

Situation

In recharge service platform, I was assigned a new requirement to store each user specific data for future analysis. The initial approach was to get these bills from external service then validate and format the data and store it in database synchronously

Task

My responsibility was to develop this feature without any impact on performance and scalability of service

Action

While analysing the solutioning. I identify that a request will additionally require processing and validating the bill will increase the latency of this critical service and create a scalability bottleneck.

Further analysing I recommended a different approach, while using Cassandra database as it is suitable for high write throughput which aligned with our requirements. Also, I proposed asynchronous event-driven approach using Kafka which can decouple the formatting and validation of the data. Which reduce the response time of this service and make it lightweighted

Result

This solution fulfilled the business requirements to store the bills from external sources for further analysis without impacting the response time of request service

Looking back at the solutioning, it was not just implementing the solution it was improving the service for more scalable and maintainable approached. This makes the platform service for better future growth

Q 2 : Do you recall a situation where you intensely and relentlessly focused to complete a task (can be professional work or hobby) that helped you grow?

If so, please describe this experience in a paragraph or two.

Situation

During cloud migration initiative, the task was to migrate an existing service to a new cloud environment. The service was assigned which consisted of Kotlin and Java 8 service in an assigned deadline

Task

My responsibility was to migrate the service to a new cloud service with no business impact

Action

As I evaluated, the new environment services are standardised on Java 21 and Spring Boot 3. If we only migrate this service, we may end up with a service which will require additional maintenance and solutioning for each enhancement

After analysing the risk and future efforts, I proposed a solution for not just migration but upgrade of the codebase from Kotlin and Java 8 to using the standardised Java 21 and Spring Boot 3, which aligned with the configuration and pom file of other services. I worked focusing on delivery within the deadline while resolving framework migration challenges until there were no compatibility issues throughout iterative testing

Result

The project was completed in the assigned timeline. The service was aligned with the modern tech stack. This reduced the future additional maintenance and separate solutioning for enhancement

This task helps to learn about the importance to look beyond the immediate solution and focus on balancing short-term delivery with long-term maintainability

Q3 : Describe an experience of joining new team, project or environment where there was significant amount of context, information or complexity to absorb in a relatively short period of time. How did you approach learning, building clarity, prioritizing what to focus on and gradually becoming effective?

I joined large microservice environment. There were a significant service-to-service communication, business flows and architecture

To avoid being overwhelmed by this, I did it layer by layer. I started with high level approach. When the document was limited, I designed the diagram of service-to-service communication

Next step was to learn service by service; I focused on service that were required for my task. I traced a request through all the service, databases to understand how the platform worked end to end. Once that was done, I communicated with my teammates to validate and fill the knowledge gaps

This help to understand the system and start working on developing new features and deployment. Later I was able to resolve production issues. This experience taught me that becoming effective in a complex environment and understanding a business logic is more efficient that understanding everything upfront